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A BP Neural Network Prediction Model Based on Dynamic Cuckoo Search Optimization Algorithm for Industrial Equipment Fault Prediction

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ABSTRACT The fault prediction problem for modern industrial equipment is a hot topic in current research. So, this paper first proposes a dynamic cuckoo search algorithm. The algorithm improves the step size and discovery probability. Then, it introduces the change trend of fitness function value into the step size update formula to balance the search speed and accuracy. At the same time, the algorithm initial global search step is larger, while the step size of the local search is smaller in the latter part of the algorithm. In the process of discovering the global optimal solution, the probability of preserving the offspring with good fitness is increased, and the uncertainty of preference random walk is improved. As the search progresses, the probability of discovery is reduced, which makes it easy to produce new individuals in the later stage of evolution. Based on this, a BP neural network prediction model based on dynamic cuckoo search algorithm optimization is established. And the experimental results show that the proposed prediction model has faster convergence and higher accuracy.

INDEX TERMS IWSN, fault prediction, BP neural network, dynamic cuckoo search.

I. INTRODUCTION

Industrial Wireless Sensor Network has raised the economic growth point within the industrial automation in recent years [1]. Monitoring the health of industrial equipment is an important means of ensuring the safety of industrial production. Currently, wired monitoring methods are limited by the complexity of industrial environments, and wired monitoring methods are costly and difficult to scale in industrial scenarios [2]. Therefore, IWSN technology has been widely studied and applied in many industrial fields, and is expected to be a revolutionary technology [3]. It complements wired monitoring technology, expands the monitoring area, improves monitoring levels, and reduces monitoring costs [4].

The Back Propagation (BP) neural network is a multilayer feedforward network trained according to error back propagation algorithm and is one of several prediction methods, which determines the mapping relationship between the network input factors and the output (the prediction) by using historic sample data for learning and training [5], [6]. BP neural networks are self-learning, self-organizing, and adaptive, possess fast training speed, and deliver local approximation and global convergence [7]. Its learning rule is to adopt the steepest descent method in which the back propagation is used to regulate the weight value and threshold value of the network to achieve the minimum error sum of square [8], [9].

In this paper, the Dynamic Cuckoo Search algorithm (DCS) is used to optimize the BP neural network in order to avoid the BP neural network easily falling into the local minimum problem [10], [11]. At the same time, the speed of the BP neural network training convergence has improved. The ultimate goal of optimization is to obtain the optimal

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BP neural network structure, which includes the optimal initial weight and threshold of the BP neural network [12]. On this basis, a BP neural network prediction model based on DCS optimization is constructed to generate a warning for industrial equipment failure [13], [14].

II. RELATED WORKS

With the development of the modern industrial technology, the complexity and scale of equipment will become larger [15]. This development trend puts higher requirements on the reliability and operational efficiency of equipment [16]. More complex equipment structures and their components make the equipment more likely to fail and the equipment is less reliable. Therefore, during the period when the equipment is put into use, the fault prediction method is used to predict the occurrence or development trend of the fault, and timely maintenance and repair to prevent the fault from occurring which can reduce the huge economic loss caused by the equipment failure [17], [18].

In [19], the combination of genetic algorithm and fuzzy logic control algorithm was proposed. The genetic algorithm was used to predict the parameters such as quantization factor and scale factor required by the fuzzy logic control algorithm; the fuzzy logic control model of the genetic algorithm was established. The authors input the sample data of the device operation, introduced the offline learning strategy, trained the relevant parameters of the fuzzy logic control algorithm, and completed the equipment failure prediction by using the trained model. The algorithm improves the training speed of the fuzzy logic control model, but the algorithm still has certain deficiencies in the accuracy of the equipment fault prediction.

In [20], the method of signal processing of the analytical model was proposed. Based on the mathematical model of the diagnostic object, the prediction was calculated according to a specific mathematical method. The signal analysis scheme in this algorithm can penetrate the dynamic nature of the essential elements of the device system and achieve real-time predictive diagnosis. However, it lacks certain stability, and it is difficult to obtain an accurate mathematical model of the measured object (especially with complex industrial equipment).

An adaptive frequency-shifting variable-scale stochastic resonance algorithm based on a time-frequency index was proposed in [21]. The algorithm first uses the morphological denoising characteristics to denoise the signal, and then uses wavelet transformation to locate the fault and construct the signal model. The measurable signal was directly analyzed by using the constructed signal model, and then the eigenvalues such as variance, amplitude, and frequency are extracted, and the fault prediction diagnosis is based on this. The biggest advantage of this algorithm is that it can locate the fault location, but there is no outstanding advantage in the building speed of this model.

In [22], an algorithm based on a subtractive clustering method and human-computer interaction was proposed to construct an expert system for equipment fault prediction diagnosis. The system consists of a knowledge base, an inference engine, and a human-machine interface. Through the knowledge and experience of logical inference and symbol information processing equipment faults, the state identification, diagnosis, and prediction of the diagnostic object is possible. This solution solves the bottleneck problem of an expert system knowledge acquisition, and optimizes the speed of the constructing expert system. However, there are some shortcomings in the accuracy of fault prediction.

Villez *et al.* [23] proposed the combination of Kalman filtering and grey prediction. The algorithm processed and analyzed the data, and established a mathematical model to predict the value of the operating parameters of the industrial equipment at a future time; thereby, judging the equipment failure or the occurrence trend. The algorithm can process and analyze linear or nonlinear relationships between various variable parameters within the model through a large amount of historical data. However, the algorithm has many shortcomings and needs further research.

A new Markov Life Prediction Model (MLPM) was proposed in [24] to predict the state of equipment operation in real time from the perspective of mode conversion. The need for the invalid physical model established by the algorithm created a deep understanding of the working mode and operating mechanism of the device. The model has input and output. Through the input, a large number of sensor historical data was used to determine whether the state of the device operation is in a failed state; the device fault can be predicted and produce output. This solution has a great advantage in the speed of building a predictive model, but the model may have unpredictable faults when the device is used on a large scale for a long time.

In [25], the life distribution model was established by combining the Bernoulli Distribution and Poisson Distribution. After the scheme determines the distribution type of the model, it is necessary to select the key parameters of the model based on the fault data to establish a fault tree prediction model. The advantage of this model is that the fault tree structure is readable, the tree diagram is clear and easy to understand, and the logic is strong. Once the fault tree of the device is established, it is easy to find known hazard problems. However, the prediction accuracy of this model is related to the large amount of historical data recorded, so it is slightly insufficient to improve the prediction accuracy of the model.

Wang *et al.* [26] proposed a fault prediction diagnosis based on a quantitative wavelet and improved LDB algorithm. A quantitative wavelet base for analyzing faults is proposed by using common information entropy. At the same time, an optimal wavelet basis decomposition algorithm based on improved local discriminant basis theory is proposed for the problem of LDB itself. Firstly, the fault eigenvectors on the local discriminant basis space are constructed. The improved LDB algorithm is used to extract the features. Then the self-organization mapping net is used to map the features to the

high-dimensional space, and finally the faults are classified. The algorithm has obvious advantages in improving the fault recognition rate and reducing the computational complexity, but it does not consider the problem that the high-dimensional space network structure is difficult to determine.

Aguiar *et al.* [27] summarized the framework of fuzzy theory fault diagnosis, and improved the Type-2 fuzzy theory and used it in fault diagnosis. This scheme does not need to establish an accurate mathematical model; rather, it requires appropriate use of membership functions and fuzzy rules. Fuzzy inference can achieve intelligent intelligence of fuzzy fault diagnosis and prediction. However, for complex fault diagnosis and prediction systems, it is very difficult to establish correct fuzzy rules and membership functions, and it takes a long time.

In [28], the scheme of the RBF network optimized by the particle swarm optimization algorithm was proposed. The particle swarm optimization (PSO) algorithm helps to improve the learning ability of the RBF network, and effectively expands the processing ability of the RBF network for ambiguity and inconsistency. Strong knowledge of the processing ability was used to make a fault prediction diagnosis. This solution can help to quickly and efficiently set the key parameters and structure of the RBF network and improve the training speed of the predictive model. However, the algorithm does not consider the minimum error as the local minimum rather than the global minimum, so its prediction accuracy may be disturbed.

The main contributions of this study are as follows:

- We proposes a dynamic cuckoo search algorithm in order to optimize the typical BP neural network for industrial equipment fault prediction. Compared with similar optimization algorithm, DCS algorithm can effectively balance the search speed and accuracy.
- In the initial stage of DCS algorithm, the global search step is large, while in the later stage, the local search step is small. As the search progresses, the probability of discovery is reduced, which makes it easy to produce new individuals in the later stage of evolution.
- The proposed DCS algorithm is used to optimize BP neural network and an industrial equipment fault prediction model is established. This model is more effective and accurate compared with other prediction model.

III. DCS ALGORITHM

A. TRADITIONAL CS ALGORITHM

1) MATHEMATICAL MODEL OF CS ALGORITHM

Cuckoo Search (CS) was proposed in 2009. It is an emerging meta-algorithm. The authors are Professor Deb and Professor Xin-She Yang from Cambridge University. It is a process of imitating the cuckoo's reproduction, and combining the behavior of the cuckoo nest with the preference of Lévy's flight. The ultimate goal is to solve the problem of quickly finding the global optimum.

The CS replaces the bad solution with a good solution in each iteration. The cuckoo bird's egg can be seen as a new solution to replace the rest of the birds. In order to facilitate the establishment of a mathematical model of the CS, we have offered three ideal conditions:

- The way the cuckoos select the nest is completely random, and each cuckoo produces only one bird's egg in the same nest.
- The best bird's nest (solution) will be retained to the next generation.
- The probability of the host bird finding the cuckoo bird is P_a, where $P_a \in [0, 1]$, the number of nests is constant at n.

At the same time, in order to improve the diversity of the solution, if other birds discover the eggs of the cuckoo, we will replace the solution and randomly eliminate it by adopting the Lévy flight preference random walk to generate a new solution.

According to the characteristics of Lévy flight, Professor Yang of Cambridge University proposed the equation for the location update of cuckoo. The equation is shown in (1):

$$X_i^{(k+1)} = X_i^{(k)} + \alpha \oplus \text{Lévy}(\lambda) \tag{1}$$

Each generation's solution is updated on the basis of the previous generation solution, and the length of the flight path determines the change value of the solution. Where $X_i^{(k)}$ represents the K th solution of the i th generation, and the solution is the position of the bird's nest in the algorithm. α is the step factor, which is a constant greater than zero. $\oplus$ is an operator, which is called a dot product; Lévy(λ) is a mathematical expression of Levi's flight mode, which is a search path that satisfies the heavy-tailed distribution, as shown in equation (2):

$$\text{Lévy}(\lambda) = \frac{\varphi \times u}{|v|^{\frac{1}{\lambda}}}(X_i^{(k)} - X_b^{(k)}) \tag{2}$$

where $X_b^{(k)}$ represents the K th optimal solution, u and v satisfy the normal distribution.

Each iteration generates a solution, which is generated by the position update equation. The new solution is used to replace the original randomly generated solution. After the replacement, the host bird discovered the bird's egg. In this process, for each new solution, r is randomly generated, and r is a random number between 0 and 1. Compare r and the host bird to find the probability P_a of the cuckoo bird. If r is greater than P_a, then the value of $X_i^{(k)}$ will be changed by preferring random walk. On the contrary, keep its value unchanged, save the solution with better fitness function value, and record it as $X_i^{(k+1)}$. The mathematical simulation formula for this process is shown in (3):

$$X_i^{(k+1)} = \begin{cases} X_i^{(k)} + r(X_i^j - X_i^e) & r > P_a \\ X_i^{(k)} & r \le P_a \end{cases} \tag{3}$$

In the equation, X_i^j and X_i^e are the two solutions of the i th generation, which are two arbitrary solutions, representing the positions of two randomly selected bird eggs (bird's nest).

From the mathematical model of the CS, we can see that the CS is somewhat similar to the genetic algorithm. Through continuous iterations, the optimal solution is finally retained. The so-called optimal is the best fitness value. In the process of optimization, one must carefully search in a small range, short-distance bounce, and occasionally walk long distances to another range to continue searching. It avoids the algorithm falling into the local optimal solution and prevents repeated searches in the same range. Short-distance searches can improve the quality of the solution. Long-distance searches can avoid the local optimization of the solution, prevent iterative iteration in a fixed range, and improve the diversity of the solution.

2) PARAMETER SELECTION OF CS

The parameters of the CS include the probability P_a of the host bird finding the cuckoo bird, the size n of the nest and the preferred walking step factor α. The value of the step factor α is related to the size of the search range. The larger the range, the larger the step size required, which can improve the speed of searching for the optimal solution. However, in the case of a small search, an excessive step size will easily deviate from the location of the solution of the best fitness solution. In the tradition CS, the value of the step factor α is fixed, and is generally taken as 0.01.

Similarly, the speed at which the optimal solution is searched is related to the population size n and the probability P_a of the bird's egg being found, according to experimental arguments, and with reference to the vast majority of literature. The general problem is that the number n of the nest size can be optimized between [15] and [25], and the probability P_a of the bird egg is found to be [0.01, 0.5].

B. DCS ALGORITHM

1) THE BASIC IDEA OF IMPROVING THE ALGORITHM

The traditional CS algorithm combines the behavior of the cuckoo nest with the preference travel of the Lévy flight, and the Lévy flight step value is determined by the u and v that satisfy the normal distribution, so the step size is completely random. Although the random step size can satisfy the diversity of the solution and avoid the algorithm falling into the local optimal solution, the uncertainty of the size of the step will affect the convergence speed of the algorithm. If a smaller step size is used, the accuracy of the solution can be improved in a local range, but the convergence speed of the global algorithm is reduced. If a larger step size is used, the speed of the algorithm global search can be improved, but the optimal solution precision of the algorithm search becomes lower. Therefore, in order to find the balance between search accuracy and search speed, as shown in equations (4) and (5), this paper improves the formula of the position update of the traditional CS algorithm.

$$X_i^{(k+1)} = X_i^{(k)} + \alpha \frac{\varphi \times u}{|v|^{\frac{1}{\lambda}}}(X_i^{(k)} - X_b^{(k)})|\frac{f_{k+1} - f_k}{f_k}| \quad (4)$$

where $X_b^{(k)}$ represents the value of the optimal solution at the K th iteration, and f_k represents the value of the fitness function at the K th iteration.

$$\alpha = \alpha_{max} - \frac{N_{iter} - 1}{N_{max} - 1}(\alpha_{max} - \alpha_{min}) \quad (5)$$

where α_{max} is the maximum step factor, α_{min} is the minimum step factor, $\alpha_{max} = 0.5$, $\alpha_{min} = 0.01$, N_{max} is the maximum number of iterations, and N_{iter} is the current number of iterations.

The Dynamic Cuckoo Search algorithm (DCS) proposed in this paper improves the step size. The equation (4) takes the change trend of the fitness function value as a measurement target and takes into account the calculation of the position update equation. In this way, when the trend $|\frac{f_{k+1}-f_k}{f_k}|$ of the fitness function value is large, the calculated step size is also large, so that a faster search speed can be ensured to adapt to a larger change trend. When the search process of the algorithm tends to be stable, the change trend $|\frac{f_{k+1}-f_k}{f_k}|$ of the fitness function value is small at this time, and the calculated step size is also small. A small step size is used to ensure the accuracy of the local search. The equation (5) realizes that the step factor decreases as the number of iterations of the algorithm increases, while the step factor α is satisfied to be between 0.01 and 0.5. Since the current iteration number N_{iter} is taken as an influencing factor in the calculation of the step factor, it is realized that in the initial stage of the algorithm, a larger step size is adopted, which can speed up the global search. The smaller step size adopted in the later stage of the algorithm improves the accuracy of the algorithm search in the local range.

In the standard CS algorithm, a random number r is compared with the discovery probability P_a, and based on the comparison result, it is determined whether a new individual is generated. If $r > P_a$, the original individual is eliminated and a new individual is generated at the same time. If $r < P_a$, only the original individual is saved. Thus the parameter P_a can be understood as the probability that the parent individual is retained. However, in the process of algorithm optimization, as the individual's quality is gradually improved, the intensity of the population evolution should be appropriately increased to avoid the algorithm falling into the local optimal solution. A suitable P_a should be gradually reduced as the search progresses, so that it is easier to generate new individuals in the later stages of evolution. Therefore, the value of P_a in the traditional CS algorithm is improved so that the value of P_a is not fixed, as shown in equation (6).

$$P_a = P_{amax} \cos(\frac{\pi}{2} \times \frac{N_{iter} - 1}{N_{max} - 1}) + P_{amin} \quad (6)$$

where N_{iter} represents the number of current iterations, N_{max} represents the maximum number of iterations, the value of P_{amax} is 0.5, and the value of P_{amin} is 0.01. Equation (6) guarantees that the value of P_a is between 0.01 and 0.5 due to The value of N_{max} is fixed, and according to the law of decreasing cosine, it is also ensured that the probability P_a is

decreased as the number of times of the current iteration N_{iter} increases.

The traditional CS algorithm can improve the diversity of solutions by solving the mechanism of discovery, and avoid the problem that the solution falls into the local minimum. The discovered solution will be discarded, and the new solution will be generated by the location update formula in a preferred way to replace the discovered solution. This mechanism can improve the performance of the algorithm global search. But in this mechanism, for each new solution, r is randomly generated between 0 and 1. Compare r and the host bird to find the probability of the cuckoo bird's egg probability P_a. If r is larger than P_a, replace the found solution with a new solution, then keep its value unchanged, and save the solution of the fitness function value. In this way, there is a problem. Since r is completely random, there is a possibility that the solution with a good fitness function value is discarded and the solution with a bad fitness function value is retained instead, which will ultimately affect the quality of the algorithm's optimal solution and the speed of convergence. Therefore, the value of the fitness function of the solution is taken into account as a measure of the generation of the random number. As shown in the equation (7), the fitness function value of the solution is also taken into account in the measurement of whether it is found and discarded.

$$r = (r' + \frac{f_{iter} - f_{best}}{f_{worse} - f_{best}}) \div 2 \tag{7}$$

where r' is a random number between [0, 1], f_{iter} is the current fitness function value, f_{best} is the current best fitness function value, and f_{worse} is the current worst fitness function value. The r value of the solution with good fitness is preserved if it is large, while the value of the solution with poor fitness is discarded if it is small.

2) STEPS OF THE DCS

The implementation steps of the DCS are as follows:

Step 1: Use $f(x)$ to represent the objective function, where $x = (x_1, x_2, \ldots, x_n)^T$, initialize a population $x_i (i = 1, 2, 3, \ldots, n)$ with n nests, set the number of nest sizes to n, the algorithm dimension to d, the maximum step factor α_{max} to 0.5, and the minimum step factor α_{min} to 0.01. The maximum discovery probability P_{amax} is 0.5, the minimum discovery probability P_{amin} is 0.01, and the maximum number of iterations N_{max};

Step 2: Use the n solutions of $x = (x_{1,1}, x_{1,2}, \ldots, x_{1,n})^T$ to represent the nest positions randomly initialized, calculate the value of the objective function for each solution separately, and record the best solution of the fitness value with f_{best} (the nest position). At the same time, use f_{worse} to record the solution with the worst fitness value (bird's nest position);

Step 3: According to the current number of iterations N_{iter}, the step factor α is calculated by the equation (5), and the best solution (the nest position) of the previous generation fitness value is retained. The next step is calculated based on the calculated step factor and the value of the current objective function. Substitute the step size into the position update equation (4), update the remaining solutions (the position of the nest), and calculate the fitness function value corresponding to the solution (the position of the nest) after the update. Compare the best fitness values with the previous generation, and finally record the best ones (the nest position);

Step 4: For each solution (the nest position), a number r' between 0 and 1 is randomly generated, and the corresponding r' is calculated according to the equation (7), and the number N_{iter} of the current iteration is substituted into the dynamic discovery probability equation (6); thereby, calculating the probability P_a of the host bird to find the cuckoo bird. Compare the size of P_a and r. If r is greater than P_a, the corresponding solution (the nest position) is randomly changed. Conversely, do not change the position of the nest. Calculate the value of the fitness function corresponding to the solution (the nest position) after the change, compare it with the solution of the previous generation (the nest position), and finally record the solution with the best fitness value (the nest position) with $X_b^{(k)}$;

Step 5: Calculate the objective function value $f(X_b^{(k)})$ corresponding to the best solution (nest position) $X_b^{(k)}$ recorded in step 4. Determine whether the objective function value satisfies the termination condition. If it is satisfied, then $X_b^{(k)}$ is the global optimal solution of the algorithm's final search. If not, return to step 2;

Step 6: Output the best solution of the fitness value recorded in step 5 (bird's nest position) $X_b^{(k)}$, which is the global optimal solution (bird's nest position).

C. DCS PERFORMANCE SIMULATION

In order to realize the performance simulation test of the DCS proposed in this paper, the traditional cuckoo search (CS) and particle swarm optimization (PSO) are used to compare and test. Compare the iterative optimization performance of the three optimization algorithms to the same objective function. A large number of scholars have shown that finding the extreme of the objective function can test the optimization performance of the optimization algorithm. Regarding the objective function selection for testing the performance of the optimization algorithm, three test functions widely used in numerical problems can often be used, and the optimization algorithm is used to search the minimum value of the same objective function. Furthermore, the optimization performance of the three optimization algorithms is evaluated and compared. The minimum value in the three test functions of the Ackley function is relatively difficult to find because of its complexity. Therefore, the number of iterations is set to 2000 when the objective function is in Ackley function, and the number of iterations of the other two test functions is 1000. The function values are all 0. The DCS has a Nest Size of 20 and a Problem Dimension of 30.

The specific definitions of the three objective functions used to test the performance of the optimization algorithm are as follows:

(1) Sphere function

$$f(x) = \sum_{i=1}^{n} x_i^2 \tag{8}$$

The Sphere function is a simpler single-peak function, easy to optimize, and the sum of squares, the search range is −100 to 100.

(2) Rastrigin function

$$f(x) = \sum_{i=1}^{n} (x_i^2 - 10\cos(2\pi x_i) + 10) \tag{9}$$

The search range of the Rastrigin function is −50 to 50, which is a multi-peak function with more local optimal values, and it is more difficult to find the global optimal solution.

(3) Ackley function

$$f(x) = -20exp(-0.2\sqrt{\frac{1}{n}\sum_{i=1}^{n} x_i^2}) - exp(\frac{1}{n}\sum_{i=1}^{n}\cos(2\pi x_i)) + 20 + e \tag{10}$$

The Ackley function has a search range of −100 to 100. It is a multi-peak function with more local minima and it is not easy to get a global optimal solution.

The curve of the three algorithms for the test function optimization iteration is shown in the following figure:

Fig.1 to Fig.3 show the three algorithms for the optimization of the three test functions. The ordinate in the simulation graph represents the optimal value of the objective function found by the three optimization algorithms. The abscissa in the graph represents the number of iterations of the optimization algorithm. Fig.1 shows the curve of the three algorithms for the sphere function optimization iteration. The sphere function is a simple single-peak function, which is easy to optimize. It is easy to see from the simulation diagram that the DCS represented by the black line iterative convergence is faster. Fig.2 shows the curve of the three algorithms for the optimization of the Rastrigin function. It can be seen from the simulation, that the convergence speed of the three algorithms is very fast in the first 50 iterations, but only after entering the interval where the optimal solution is located. The DCS represented by the black line can quickly iterate to the optimal value of the objective function, compared to the PSO represented by the blue line and the CS represented by the red line. Fig.3 shows the curve of the three algorithms for the optimization of the Ackley function. Since the Ackley function has more local minimum values, it is not easy to obtain the global optimal solution. Therefore, the three algorithms are more iterative, from the simulation diagram. It is not difficult to see that when the optimal value of the objective function is finally found; the PSO represented by the blue line and the CS represented by the red line are the DCS iteration represented by the black line more than twice the number of times. According to the performance simulation

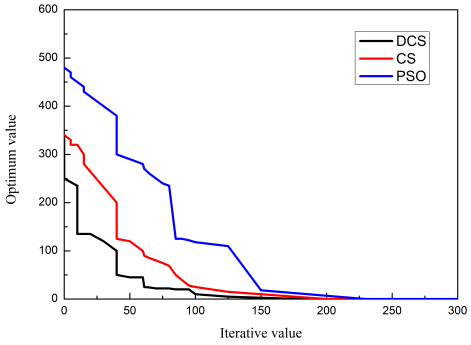


FIGURE 1. Three algorithms for the Sphere function optimization curve.

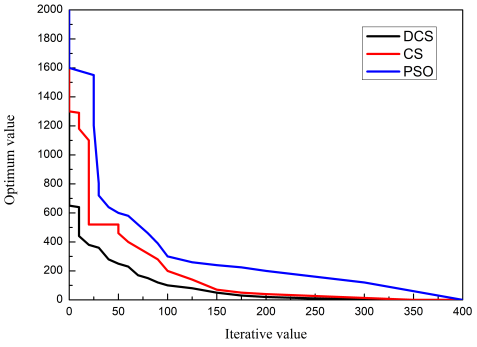


FIGURE 2. Three algorithms for the Rastrigin function optimization curve.

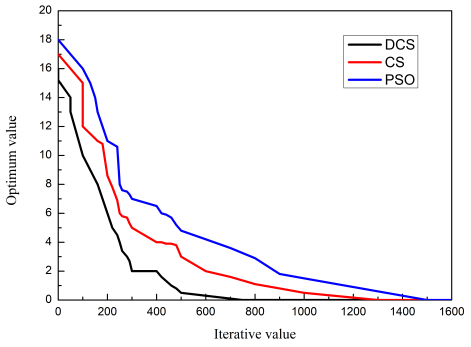


FIGURE 3. Three algorithms for the Ackley function optimization curve.

test of the three algorithms, the DCS is used to deal with the optimization problem when comparing the optimization process of the same test function with the traditional two representative optimization algorithms. With obvious advantages, the DCS finds the objective function global optimal solution with fewer iterations and the convergence iteration faster.

Table 1 represents the objective function optimization test results of DCS, CS and PSO, and records three optimization algorithms for three test functions, including the obtained variance, the optimal objective function value, and the worst objective function value. By comparing the data in the table, we can analyze that the optimal objective function found by the DCS is compared with the CS and PSO for the same test function. The smaller value is closer to 0, which shows

TABLE 1. Objective function optimization test results of DCS, CS and PSO.

Function	Algorithm	Optimal value	Worst value	Variance
Sphere	PSO	9.8742E-03	9.9897E-01	4.4296E-03
	CS	7.7432E-06	8.2215E-05	2.9068E-10
	DCS	4.2521E-13	1.6324E-11	1.73875E-23
Rastrigin	PSO	6.8239E-02	3.4298E-01	4.56232E-06
	CS	5.4233E-03	2.8367E-02	5.18217E-05
	DCS	3.5621E-06	1.0125E-04	6.29377E-10
Ackley	PSO	4.7365E-02	3.7895E-01	5.36872E-06
	CS	3.2125E-04	2.5863E-03	3.49616E-07
	DCS	2.6849E-07	1.6742E-06	1.12949E-13

that the DCS has better optimization ability, and the accuracy of the optimal objective function value found is higher. In addition, for the same test function, the variance of the DCS is also significantly smaller than the other two optimization algorithms. The smaller the variance, the better the stability of the algorithm.

IV. BP NEURAL NETWORK PREDICTION MODEL BASED ON DCS OPTIMIZATION (DCS-BP)

The BP neural network is optimized by the MetaHeuristic Algorithm, and the analysis is essentially an optimization of the BP neural network structure. In the process of optimizing the BP neural network by the MetaHeuristic Algorithm, the fitness function can be regarded as the structure of the BP neural network. The ultimate goal of optimization is to obtain the optimal solution of the BP neural network structure. By using the DCS to optimize the BP neural network, not only can prevent the BP neural network from falling into the local minimum problem, but also the speed of BP neural network training convergence can be improved. The ultimate goal of optimization is to obtain the optimal BP neural network structure, such that it meets the optimal initial weight and threshold of the BP neural network. The optimal BP neural network can be used to better realize a warning for industrial equipment failure.

In the whole process of the DCS-BP algorithm, the nest (solution) obtained by DCS optimization represents the initial optimal weight and the threshold of the BP neural network structure, and iteratively optimizes each layer of neurons in the BP neural network through iterative iteration. The weights and thresholds of the connections are obtained, and the structure of the optimal BP neural network is obtained after the assignment. By inputting the training samples to the obtained optimal BP neural network, the repeated learning iterations of the neural network finally make the error of the output value and the expected values reach the actual problem. At the same time, the DCS will continue to optimize the BP neural network. The structure continues to work until the conditions for stopping are met.

The DCS is used to optimize the parameters of the prediction model. Compared with other traditional optimization algorithms, it is more efficient in optimizing parameters and getting global optimal solutions faster. The main reasons are twofold:

(1) The algorithm has good efficiency. The whole algorithm is a process of optimization. In the process of optimization, the most efficient path of optimization is adopted, and a balance point between global search and local search is found.

(2) The algorithm has good applicability, because there are only two key parameters of the CS algorithm. The two key parameters are the probability of bird egg discovery and the size of the population.

The training process of the BP neural network is essentially gradient-decreasing. It has been analyzed in the past that it is prone to fall into the local minimum. The initial weights and thresholds of the connections between the neurons in the BP neural network range from -1 to 1. The selection of the initial weight and threshold of the BP neural network is very important to the learning process. If the initial value is too large or too small, the training speed will be too slow, and the predicted effect of the trained model is not good. In the third chapter, we improve the traditional optimization algorithm CS algorithm, and propose a DCS. The DCS has a fast convergence speed, and the optimal solution has a high precision. There are obvious advantages when it comes to problems. We can optimize the structure of BP neural network according to the DCS, that is, optimize the optimal initial weight and threshold to accelerate the speed of training the BP neural network. Finally, the BP neural network is best through training. The results of predicting industrial equipment failures using the optimal BP neural network after learning are more accurate. The pseudo code of the DCS-BP algorithm is as follows:

V. SIMULATION AND ANALYSIS

By applying the algorithm to the actual industrial equipment fault prediction, and taking the temperature parameters of the electromechanical equipment as an example.

(1) Set the number and location of sensor nodes to construct an industrial wireless sensor network, set the time period of the acquisition, transmit the temperature data of the device operation to the server during the set period, and finally store it in the database of the server.

(2) Design the DCS-BP network, pre-process the historical temperature data of this industrial equipment by obtaining the historical data stored in the database, and finally obtain the training samples and test samples.

(3) Train the DCS-BP network through the obtained training samples, and finally obtain the trained DCS-BP network.

(4) Record the network structure after training, such as the weight and threshold of the connections between the neurons in each layer, and test the DCS-BP network by using the test samples obtained from the server database and by analyzing the test values and actual values. The error between the values is used to determine whether the trained DCS-BP network meets the actual accuracy requirements.

(5) Finally, the network model that can meet the requirements of industrial equipment fault prediction accuracy is stored and recorded, and the model is used in the industrial

Algorithm 1 DCS-BP Algorithm

Begin:

1: Determine the topology of the BP network, initialize the BP network weights, and thresholds;

2: DCS encodes the initial weights and thresholds of the BP network;

3: Input sample data, then pre-processing sample data;

4: Calculating the mean square error after the BP network training, recorded as fitness value f;

5: **While** (The fitness value f does not meet the accuracy requirements) **do**

6: According to DCS location update formula, calculating fitness values;

7: Select, replace, eliminate;

8: Get the optimal initial weight and threshold of the BP network;

9: **While** (Maximum number of iterations not reached) **do**

10: Optimized BP network calculation error for training samples;

11: Update the weight and threshold of the BP network;

12: **End While**

13: **End While**

14: Get the best BP neural network optimized by DCS, output prediction structure;

End

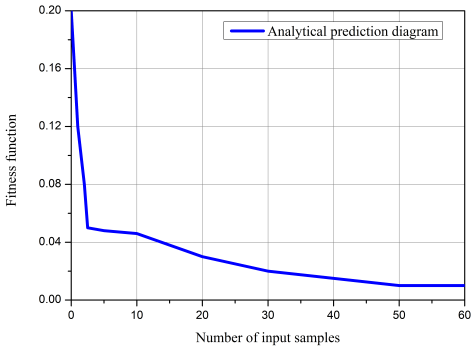


FIGURE 4. BP neural network based on dynamic cuckoo algorithm optimization training process diagram.

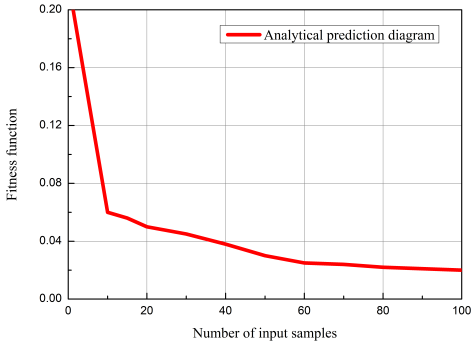


FIGURE 5. Traditional BP neural network training process diagram.

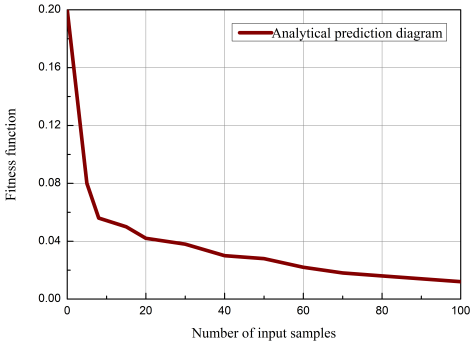


FIGURE 6. BP neural network based on particle swarm optimization training process diagram.

equipment monitoring system to complete the system fault prediction function.

The IWSN is used to monitor the electromechanical equipment to obtain real-time data of the equipment operation. Taking the temperature data of the equipment as an example, 1000 sets of data are selected and the data are randomly divided into two groups according to a 4:1 ratio. One group is training sample A and the other group is test sample B. Through training of training sample A, the BP neural network model based on particle swarm optimization algorithm, DCS-BP network, and original BP neural network can be obtained. Then, using sample B, the three models established are verified, and the results of the three models are compared and analyzed.

Through a large number of experimental simulation results, the network in this paper chooses the structure of 13-11-1, and the training method of the BP neural network is a dynamic gradient. The training of the entire model is done in Matlab, calling the toolbox of its neural network. The training function is the Traingd function, the hidden layer uses the Tansig function, the output layer uses the Purelin function, and the maximum number of iterations is set to 300. The fitness value calculated after the final training is less than 0.01 to terminate the training. The population size n in the CS is set to 10, and the maximum number of iterations N_{max} is set to 10, and the dimension thereof can be calculated as $d = 13 \times 11 + 11 \times 1 + 12 = 166$. The training result of DCS-BP is shown in Fig.4:

According to the above initial parameters, the traditional BP neural network model and the BP neural network model based on particle swarm optimization are established. The simulation results are shown in Fig.5 and Fig.6.

Test three samples were used to verify the three models. After the training and verification of the above three models, the actual and predicted values of the three models are shown in Fig.7 and Fig.8 and Fig.9, respectively.

The performance comparison table of the three models is shown in Table 2. The root mean square error of the

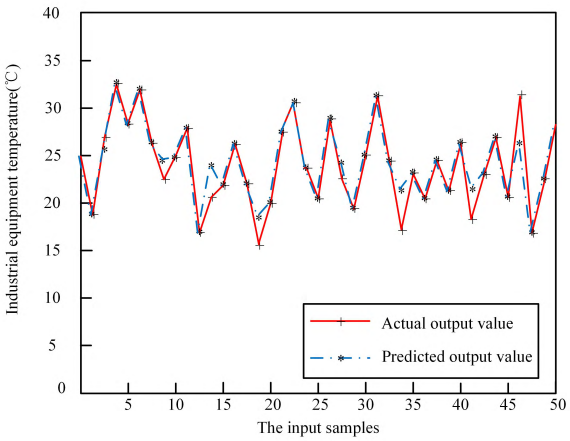


FIGURE 7. DCS-BP model verification diagram.

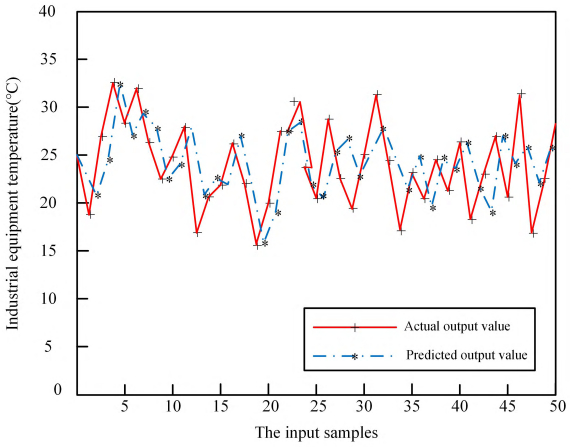


FIGURE 8. Traditional BP neural network model verification diagram.

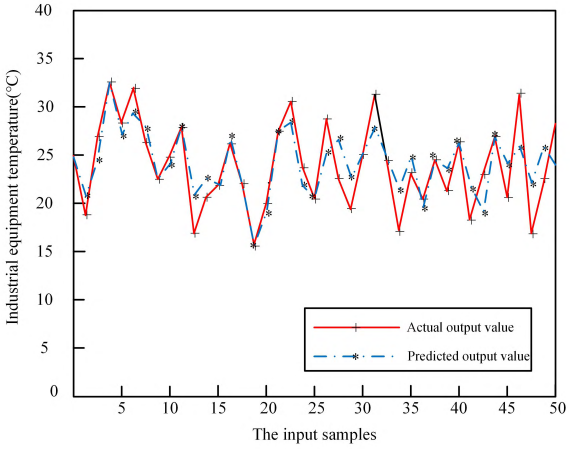


FIGURE 9. BP neural network based on particle swarm optimization model verification diagram.

three models and the average absolute percentage error are compared by calculation.

It can be seen from Table 2 that although all three models can complete the prediction task of industrial equipment

TABLE 2. Three model performance comparison table.

Model category	Sample set	Number of hidden layer nodes	Average absolute percentage error	root-mea n-square error
Traditional BP neural network	A	11	0.007394	3.045×10^{-7}
	B		0.004930	0.003624
BP neural network based on particle swarm optimization	A	11	0.006478	2.216×10^{-7}
	B		0.003562	0.002768
BP neural network based on DCS algorithm optimization	A	11	0.005635	1.528×10^{-7}
	B		0.002245	0.001973

failure, the overall performance of the BP neural network based on the dynamic cuckoo algorithm is better than the traditional BP neural network and particle swarm optimization. The BP neural network is good, the training speed is faster, and the prediction accuracy is higher.

In actual operation, relying solely on the operating device parameters, such as the temperature of the forecasting device to determine whether the industrial device has a potential fault depends on the subjective awareness of the device administrator. Additionally, the BP neural network based on the dynamic cuckoo search algorithm cannot guarantee that all predicted values are highly consistent with the actual value, it can only guarantee that the global error is the smallest. At the same time, there is interference at the site where the industrial equipment is operating, so there will be a small amount of predicted values that are far from the actual value. Given these realities, we need to find a mechanism for an early warning of failures to help device administrators, rather than relying solely on the subjective judgment of device administrators.

The error between the predicted and actual values here is as follows:

$$\varepsilon_n = x_{nest} - x_n \tag{11}$$

where x_{nest} is the prediction value and x_n is the actual value.

The fault prediction model is established based on the sample data, and the sample data used for training is basically derived from the normal operation of the device. The sample data amount is large enough to cover the normal working range of the industrial equipment. Therefore, this model will work as long as the equipment does not malfunction, the error fluctuation calculated by the trained network is within a certain range, and there is a small amount of data such that the error exceeds the maximum value due to the low precision of the fitting. However, these data will not form an increasing trend of error, and the error of the subsequent data will return to a reasonable interval.

If the device has a tendency to fail, the error between the predicted value calculated by the network after training and

the corresponding actual value will exceed the reasonable range, and the error will continue to increase as the equipment failure increases. Once this happens, the industrial equipment administrator can judge the equipment according to the increasing trend of the error and issue an early warning to the equipment.

VI. CONCLUSION

Aiming at the problem that the traditional BP neural network prediction algorithm has a slow convergence speed and is easy to fall into the local minimum value, this paper proposes to use the Dynamic Cuckoo Search algorithm to optimize the BP neural network structure. Firstly, based on the analysis of traditional Cuckoo Search optimization algorithm, this paper proposes a Dynamic Cuckoo Search algorithm. The algorithm balances the search speed and accuracy by improving the step size and finding the probability and applying the fitness function value to the step update equation. At the same time, in the process of discovering the global optimal solution, the uncertainty of preference random walk is improved by improving the retention probability of the offspring with good fitness. It also promoted the emergence of new individuals in the late stage of evolution. On this basis, a BP neural network prediction model based on Dynamic Cuckoo Search optimization is established. The simulation structure verifies the correctness and effectiveness of the prediction model.

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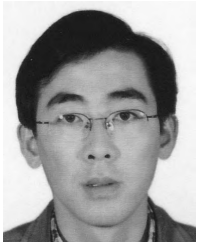
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